Ambient Systems

Machine Learning Technology for Smart Buildings

Problem Statement

Providing a healthy and safe living space depends significantly on indoor air quality, but there are currently limited ways to measure and improve it.

- Impacts comfort and productivity
- Correlated with humidity
- Energy intensive

Temperature



- Must be within a specific healthy range
- Correlated with temperature
- Impacts spread of contaminants

Humidity



- Organic: pollen, mold, bacteria
- Inorganic: VOCs, SVOCs, Ozone (O_3)
- CO_2
- Other airborne pathogens

Contaminants



Ambient Systems

Ambient is a one stop shop for building owners and operators who want to manage indoor environment and air quality seamlessly and effectively.

Building Components

- Mechanical
- Electrical
- Sensors & Controls
- Ambient Sensors*
- Ambient Controls*

Ambient Platform**

Understand
Track relevant parameters based on unique building operations

Optimize
Fine-tune building systems to achieve desired goals

Improve
Identify opportunities for technology and equipment upgrades

Applications

Healthcare
Minimize airborne infections

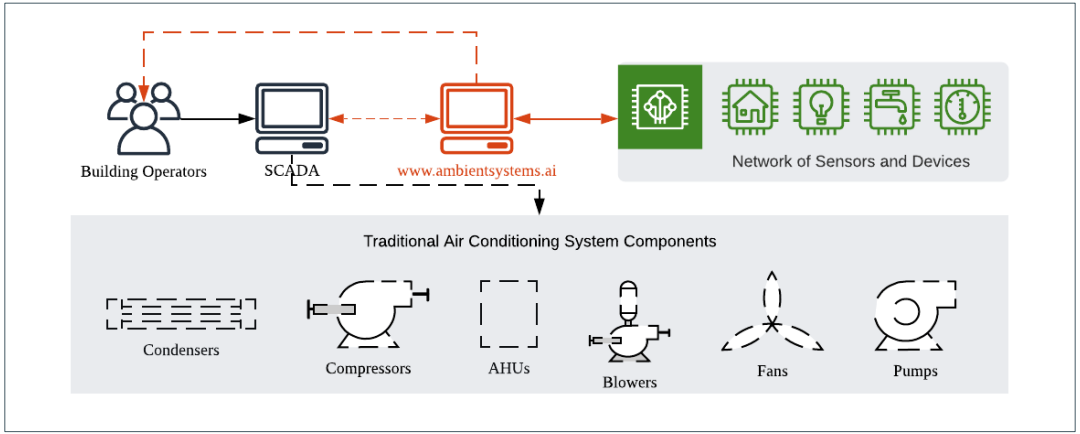
Offices
Increase worker comfort and productivity

Museums
Preserve invaluable artifacts

Vertical Farms
Increase crop yield

* Based on unique building needs. General input parameters: temperature, relative humidity, CO_2, contaminants, and user feedback.
** Platform outputs are available online, delivered by email/message notifications, or connected to your building's SCADA system

Ambient Systems
High-level Schematic



What makes Ambient different?

- Technology tailored for unique building operations and occupant needs
- Seamlessly connects to building's existing SCADA system and provides notifications directly to the operators
- Adds an additional layer of intelligence using AI-enabled optimization models
- Managed on the cloud and available across your devices
- Enabled by a network of interconnected sensors and devices (Internet-of-Things)
- Leverages NUS IP to minimize maintenance costs of Sensors and Devices using battery-less modules

Solution: Smart Buildings Product

1 Monitoring

Ambient monitors 3 key parameters through a collection of sensors and user feedback

1. Temperature
2. Humidity
3. Contaminants



Sensors + :

2 Recommendations

Ambient provides data-driven recommendations across 3 categories

1. Change temperature
2. Change humidity
3. Increase airflow



Sensors + : Recommendations

3 Automation

Ambient manages 3 groups of devices to automatically improve air quality

1. Thermometers
2. Humidifiers
3. Fans and Ventilators



Sensors + : Actuators

Recommendations
Automation

4 Machine Intelligence

Why now?

1. Increase in IoT devices such as sensors and actuators
2. Increase adoption of cloud computing
3. IoT applications have mainly focused on analytics for reporting (dashboards)
4. Adopted in manufacturing but not in buildings industry
5. Building automation techniques have focused on hardware and firmware innovations (equipment in many cases is unutilized and under-optimized)
6. New cutting-edge solutions are developed for mainly new office buildings (overlooking retrofits and non-office commercial buildings)
7. Current process for building management is very traditional: engineer comes in to audit the building and provides recommendations (lack of real-time feedback, lack of in-house expertise to manage sophisticated systems). Example, should you raise or lower humidity, temperature, or increase ventilation to address an increase in CO2 levels or particulate matter? Which combination of the three?
8. Limited Analytics and AI capabilities in-house
9. Models require adaptation for needs of building owners, operators, and occupants

Growth

Market Size

Competition

Equipment Manufacturers	Cloud Service Providers (IoT Platforms)	System Integrators (Consulting Firms)
• Siemens (Mindsphere IoT platform) • (EcoStruxure IoT platform built on Microsoft Azure) • ABB • Bosch • Rockwell • Honeywell • • Carriers	• Google (GCP) • Amazon Web Service (AWS) • Microsoft Azure • Cisco IoT • Salesforce IoT • Oracle IoT • Alibaba IoT	• Accenture • Deloitte • Capgemini • Infosys • Sada Systems • McKinsey Digital

Product

Building Type	Use Case
All building types	Improve energy efficiency
Offices and condos	Improve comfort
Offices	Improve productivity
Indoor farming	Increase yield
Museums	Protect artifacts
Industry	Prolong equipment life

Business Model

Team

Ivan Damnjanovic

- Led product development for international clean energy startups backed by Y Combinator, , and Audi
- Experienced in design of engineering systems and digital technology integration
- MComp, Computer Science, NUS (in-progress)
- BASc, Civil Engineering, U of T (2015)

Financials

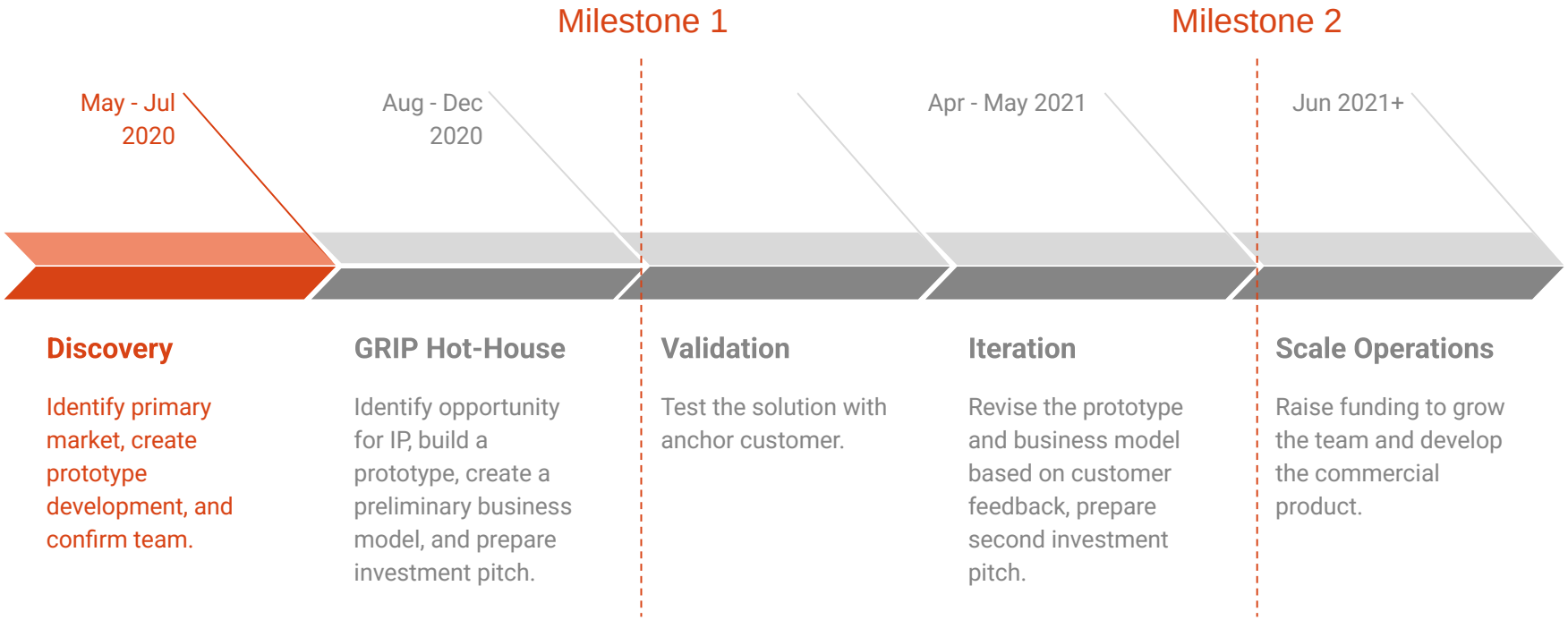
The Ask

Thank You

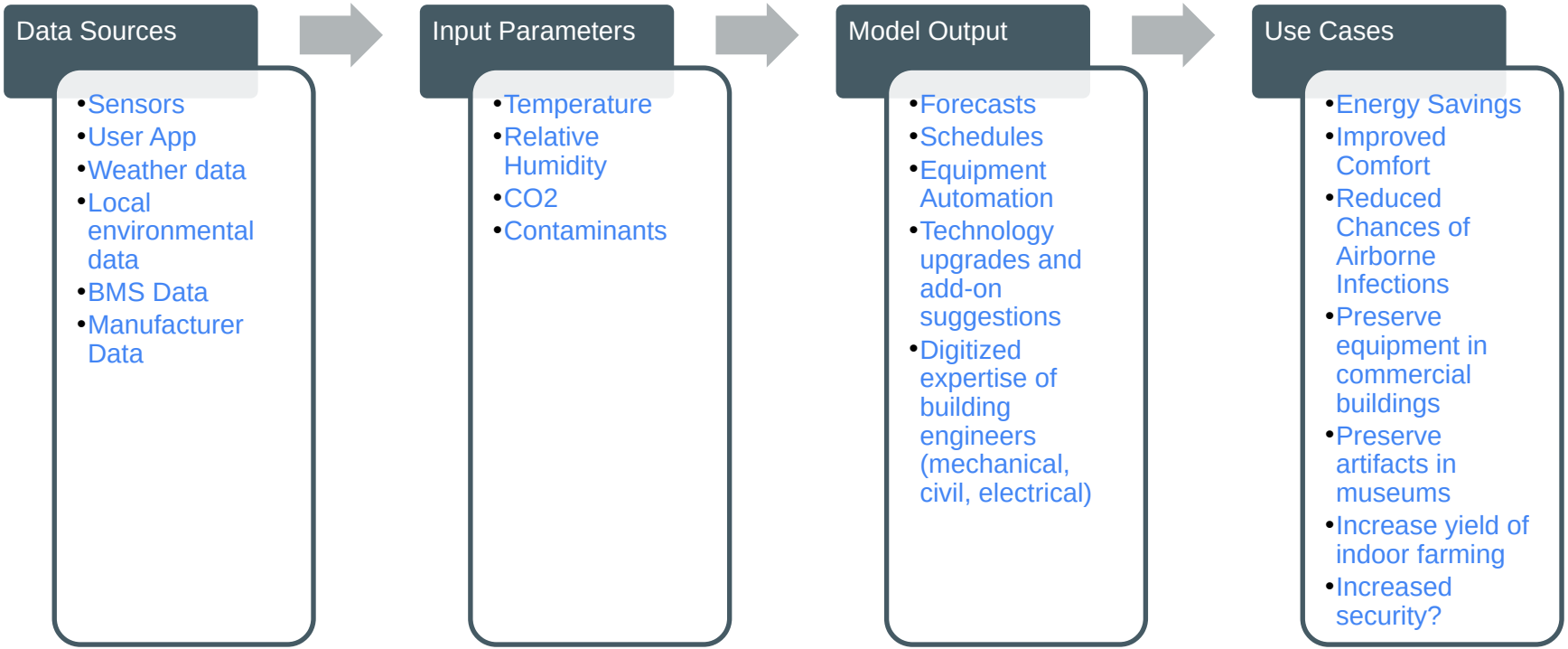
Contact

Appendix

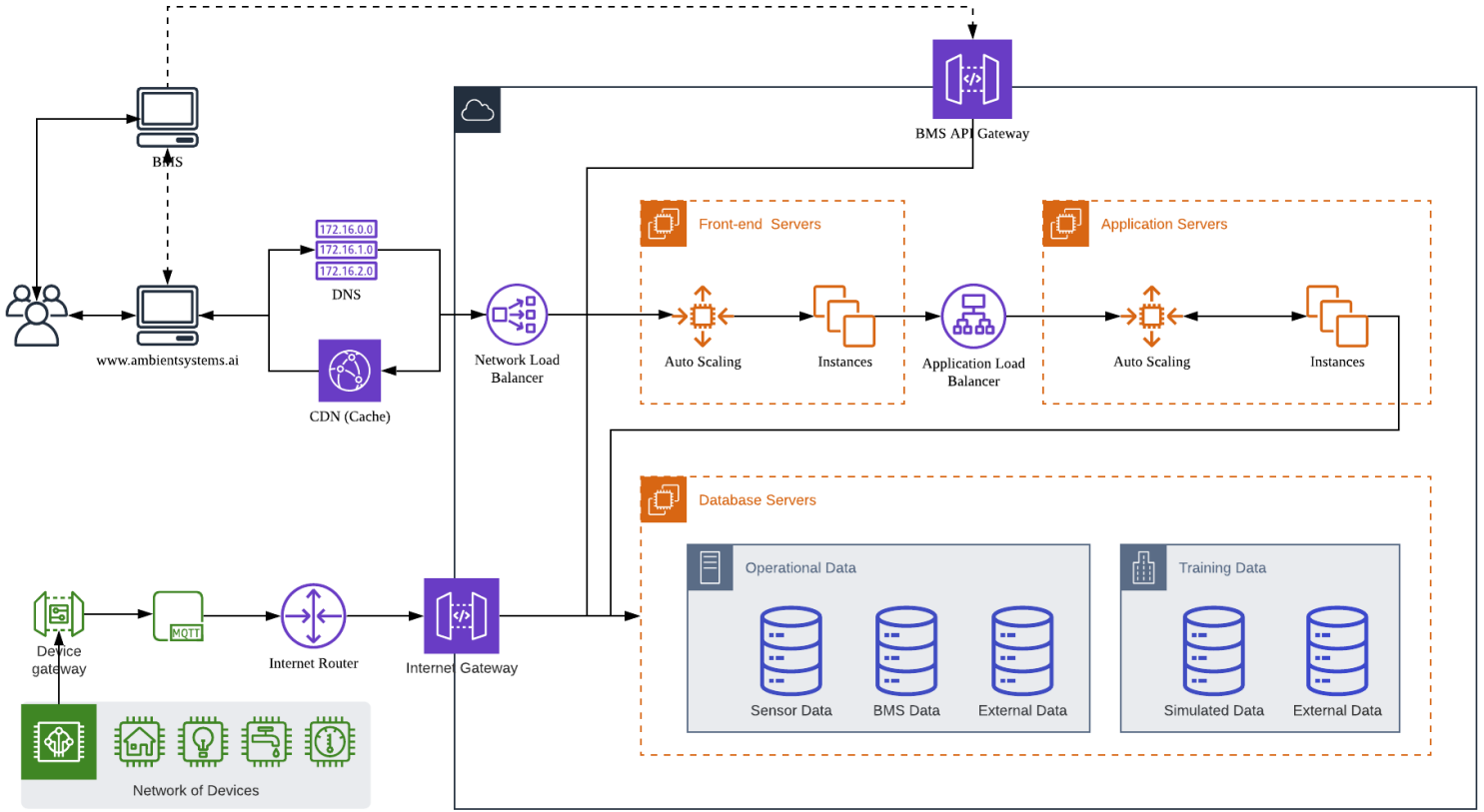
Roadmap


Milestone 1
Milestone 2
May - Jul 2020
Aug - Dec 2020
Apr - May 2021
Jun 2021+
Discovery
Identify primary market, create prototype development, and confirm team.
GRIP Hot-House
Identify opportunity for IP, build a prototype, create a preliminary business model, and prepare investment pitch.
Validation
Test the solution with anchor customer.
Iteration
Revise the prototype and business model based on customer feedback, prepare second investment pitch.
Scale Operations
Raise funding to grow the team and develop the commercial product.

Process


Data Sources
•Sensors
•User App
•Weather data
•Local environmental data
•BMS Data
•Manufacturer Data
Input Parameters
•Temperature
•Relative Humidity
•CO2
•Contaminants
Model Output
•Forecasts
•Schedules
•Equipment Automation
•Technology upgrades and add-on suggestions
•Digitized expertise of building engineers (mechanical, civil, electrical)
Use Cases
•Energy Savings
•Improved Comfort
•Reduced Chances of Airborne Infections
•Preserve equipment in commercial buildings
•Preserve artifacts in museums
•Increase yield of indoor farming
•Increased security?

System Architecture



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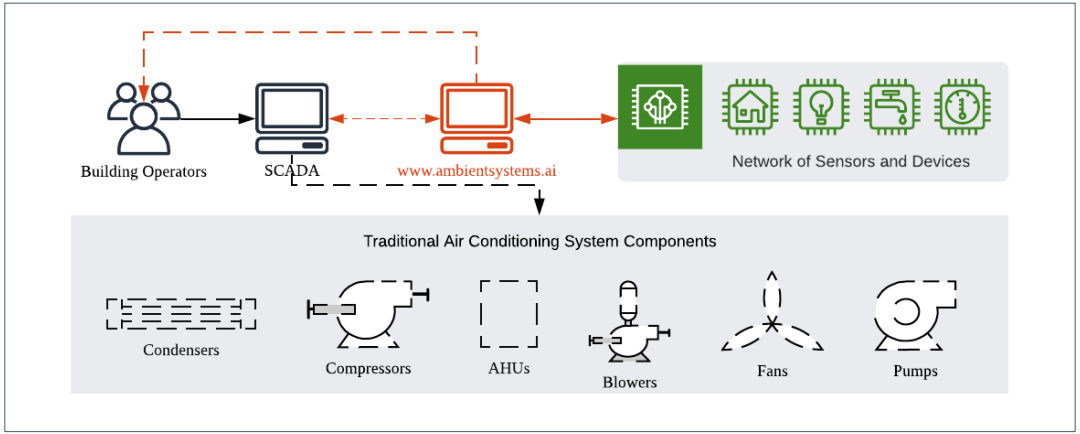
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